

Setting Up My Home Cyber Security

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Purpose

Create a safe, isolated environment (Kali attacker + Windows target) to practice reconnaissance, enumeration, exploitation (lab-only), and defensive detection.

Tools

- VirtualBox (host)
 - Windows 10 Home (victim) — Guest VM
 - Kali Linux (attacker) — Guest VM
-

Topology & Safety rules (must do first)

1. **Network:** Use a Host-Only or Internal Network in VirtualBox so VMs communicate only with each other and your host, not the Internet.
 - VirtualBox: VM → Settings → Network → Adapter 1 → Host-only Adapter (pick the same host-only for both VMs).
 2. **Snapshots:** Take a snapshot of each VM *before* every exercise so you can revert.
 - VirtualBox: Right-click VM → Snapshots → Take Snapshot.
 3. **No external targets:** Only test systems you own. Never scan or attack external machines.
 4. **Logs & notes:** Keep a lab notebook (commands + outputs + goals) for each exercise.
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Initial VM configuration

Windows 10 (victim)

- Install latest OS updates (in real-world op, but in lab you may keep intentionally vulnerable builds).
- **Create a low-privilege test account labuser:**
net user labuser P@ssw0rd123 /add
- **Create and share a folder for transfers:**
New-Item -ItemType Directory -Path C:\Shared -Force

NB: This is For Educational Purpose Only

```
& icacls 'C:\Shared' /grant 'labuser:(OI)(CI)F' /T
net share Shared=C:\Shared /grant:labuser,full
```

- **Ensure SMB2/SMB3 enabled and Samba server running:**

usually on by default; ensure LanmanServer service running

```
Start-Service -Name LanmanServer
```

```
Get-SmbServerConfiguration | Format-List
```

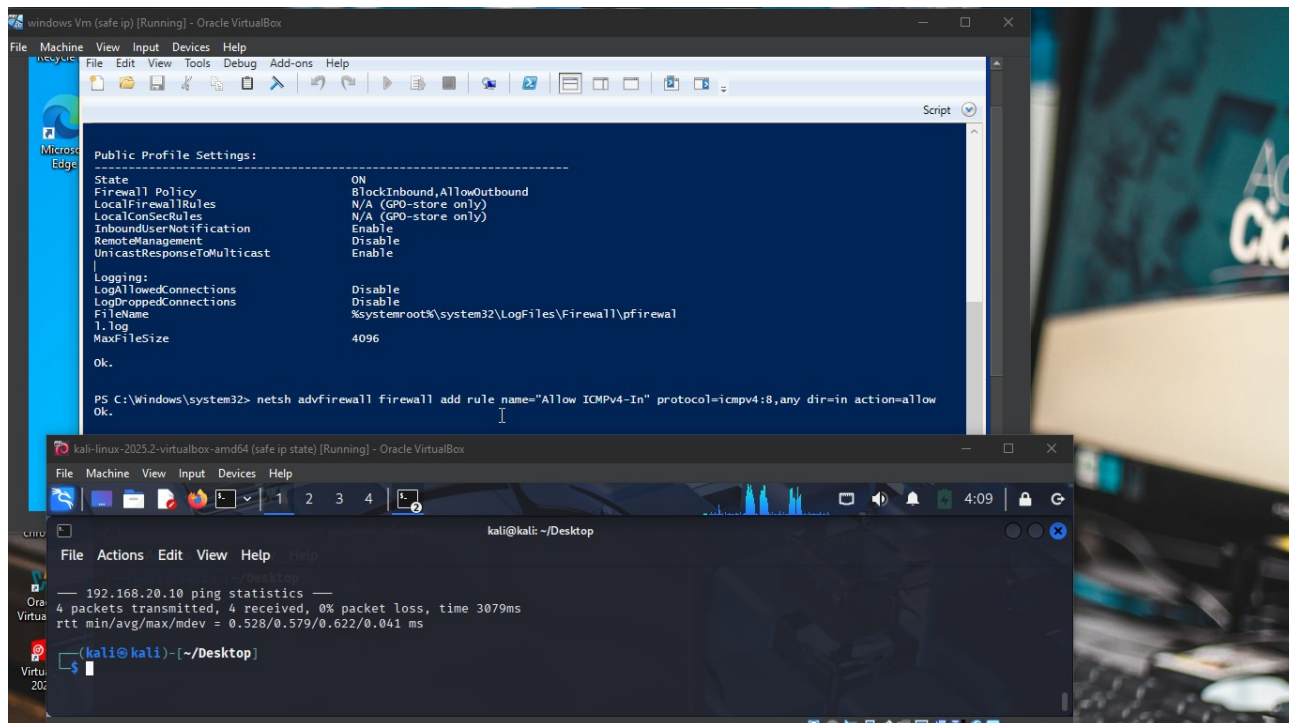
```
EnableSMB1Protocol,EnableSMB2Protocol,RequireSecuritySignature,EnableSecuritySignature
```

- **Set network to Private (lab friendlier):**

```
Set-NetConnectionProfile -InterfaceAlias "Ethernet" -NetworkCategory Private
```

```
Enable-NetFirewallRule -DisplayGroup "File and Printer Sharing"
```

- **(Or add explicit rule:)**



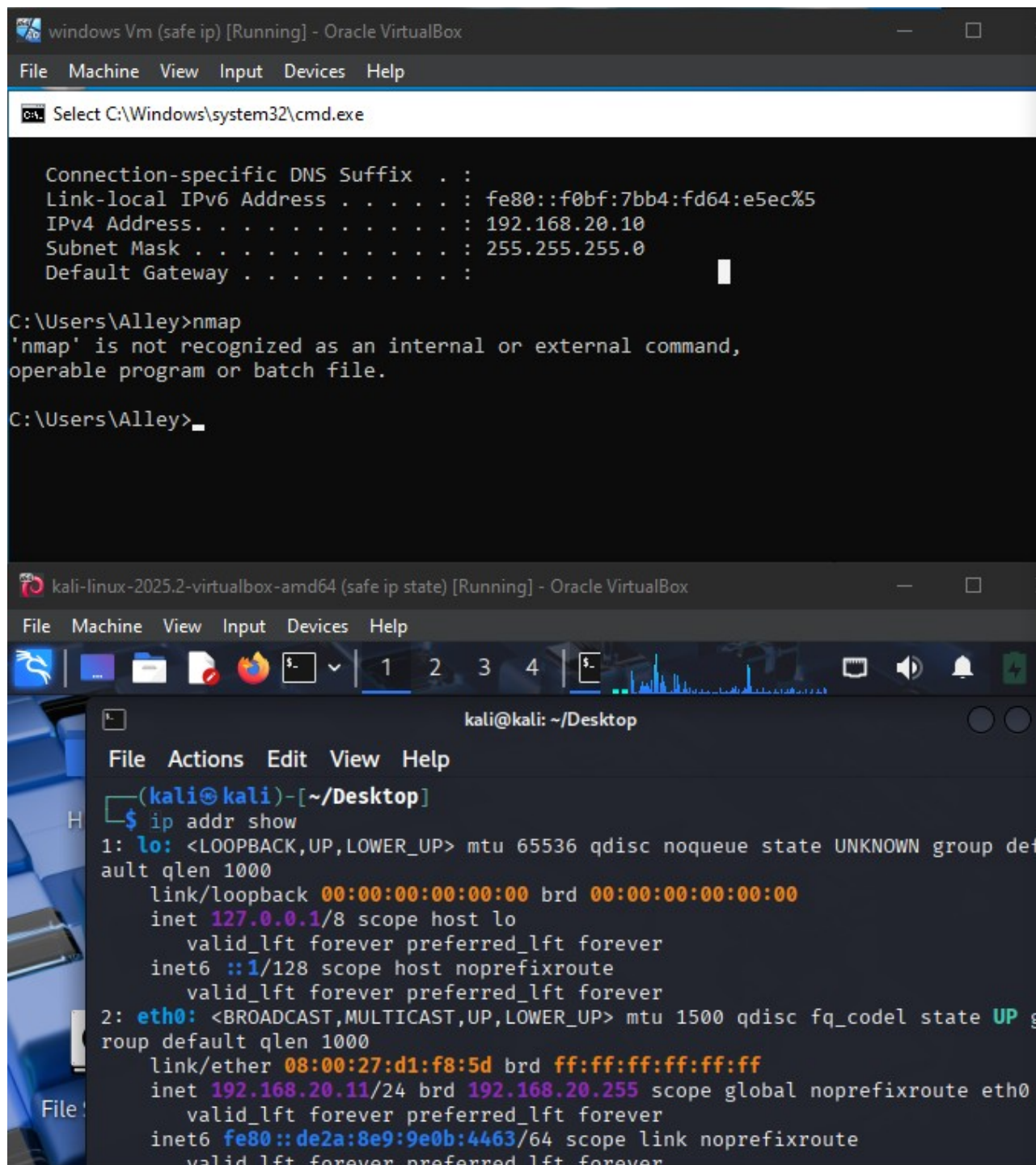
```
netsh advfirewall firewall add rule name="Allow SMB (139,445)" dir=in action=allow
protocol=TCP localport=139,445
```

Kali (attacker)

- Ensure on same host-only network and note IP:

```
ip addr show
```

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Quick connectivity verification

On Kali (replace with your IPs)

```
# ping victim
ping -c 4 192.168.20.10
```

```
# quick TCP checks
nc -vz 192.168.20.10 139 445 3389 80 5985
```

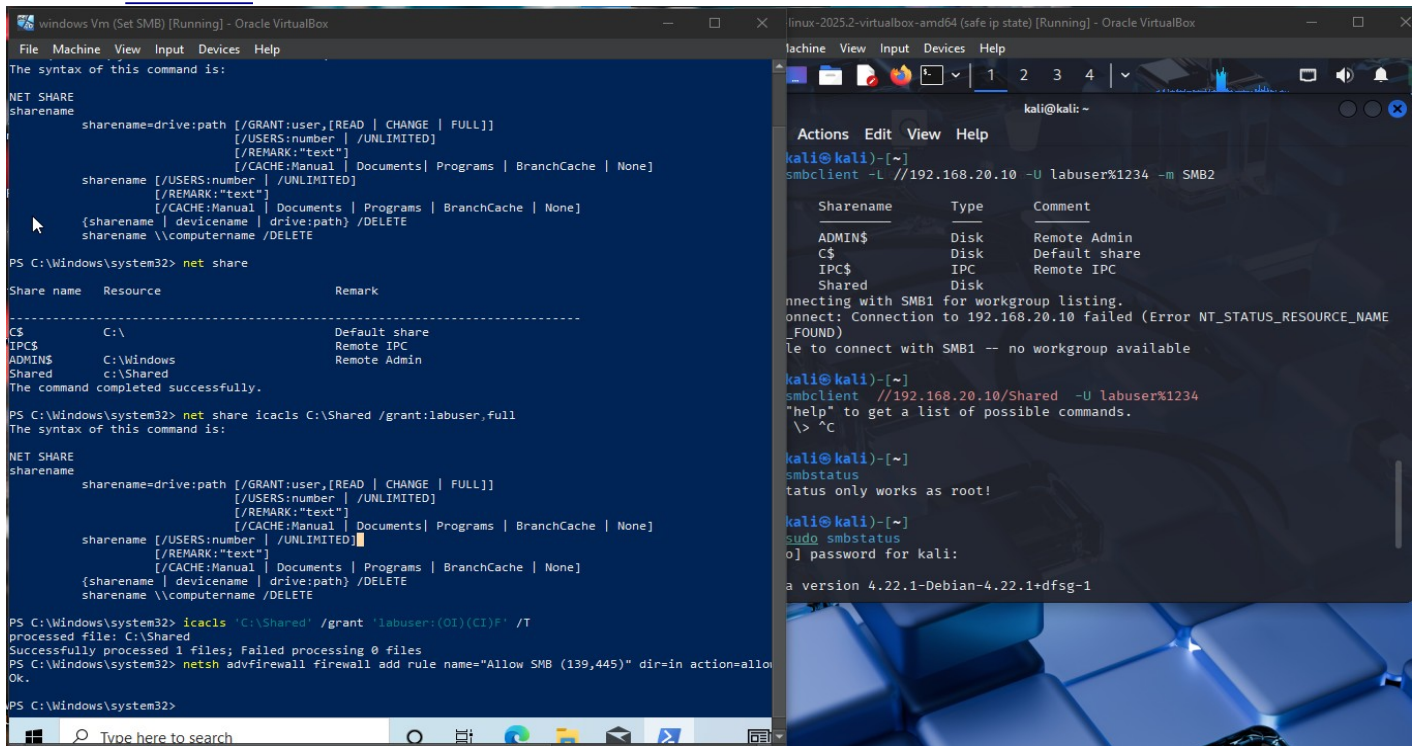
```
# list SMB shares forcing SMB3 (use credentials below)
smbclient -L //192.168.20.10 -U labuser%P@ssw0rd123 -m SMB3
```

On Windows (PowerShell Admin)

```
# confirm LanmanServer listening on 445
netstat -an | findstr LISTEN
```

```
# confirm share exists
net share
```

```
# check ACLs
icacls C:\Shared
```



```
windows Vm (Set SMB) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
The syntax of this command is:
NET SHARE
sharename
sharename=drive:path [/GRANT:user,[READ | CHANGE | FULL]]
[/USERS:number | /UNLIMITED]
[/REMARK:"text"]
[/CACHE:Manual | Documents | Programs | BranchCache | None]
sharename [/USERS:number | /UNLIMITED]
[/REMARK:"text"]
[/CACHE:Manual | Documents | Programs | BranchCache | None]
{sharename | devicename | drive:path} /DELETE
sharename \\computername /DELETE

PS C:\Windows\system32> net share

Share name Resource Remark
-----
C$ C:\ Default share
IPC$ C:\Windows Remote IPC
ADMIN$ C:\Windows Remote Admin
Shared c:\Shared Remote Admin
The command completed successfully.

PS C:\Windows\system32> net share icacls C:\Shared /grant:labuser,full
The syntax of this command is:
NET SHARE
sharename
sharename=drive:path [/GRANT:user,[READ | CHANGE | FULL]]
[/USERS:number | /UNLIMITED]
[/REMARK:"text"]
[/CACHE:Manual | Documents | Programs | BranchCache | None]
sharename [/USERS:number | /UNLIMITED]
[/REMARK:"text"]
[/CACHE:Manual | Documents | Programs | BranchCache | None]
{sharename | devicename | drive:path} /DELETE
sharename \\computername /DELETE

PS C:\Windows\system32> icacls 'C:\Shared' /grant 'labuser:(OI)(CI)F' /T
processed file: C:\Shared
Successfully processed 1 files; Failed processing 0 files
PS C:\Windows\system32> netsh advfirewall firewall add rule name="Allow SMB (139,445)" dir=in action=allow
OK.

PS C:\Windows\system32>

linux-2025.2-virtualbox-amd64 (safe ip state) [Running] - Oracle VirtualBox
Machine View Input Devices Help
kali@kali: ~
Actions Edit View Help
kali@kali)-[~]
smbclient -L //192.168.20.10 -U labuser%P@ssw0rd123 -m SMB3

Sharename Type Comment
-----
ADMIN$ Disk Remote Admin
C$ Disk Default share
IPC$ IPC Remote IPC
Shared Disk

Connecting with SMB1 for workgroup listing.
connect: Connection to 192.168.20.10 failed (Error NT_STATUS_RESOURCE_NAME
_FOUND)
le to connect with SMB1 -- no workgroup available

kali@kali)-[~]
smbclient //192.168.20.10/Shared -U labuser%P@ssw0rd123
"help" to get a list of possible commands.
\> ^C

kali@kali)-[~]
smbstatus
tatus only works as root!

kali@kali)-[~]
sudo smbstatus
o] password for kali:

a version 4.22.1-Debian-4.22.1+dfsg-1
```

On the next series I Will explore Privilege escalation using winPEAS.exe

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